MILITARY SPECIFICATION

RETAINER ASSEMBLY, SLIDING

This specification is approved for use within the USA Belvoir Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of sliding retainer assembly for connecting a treadway-type superstructure to the universal transom.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel and Seals.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.

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MIL-T-704	- Treatment and Painting of Materiel.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoi Research, Development, and Engineering Center, ATTN: STRBE-DS, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWING

D13207E9448	- Retainer Assembly, Sliding.

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN WELDING SOCIETY (AWS)

A5.1 - Mild Steel Covered Arc-Welding Electrodes.
A5.18 - Mild Steel Electrodes for Gas Metal-Arc Welding.
D1.1 - Structural Welding Code, Section 5, Qualification.

(Application for copies should be addressed to the American Welding Society, Inc., 2501 NW 7th Street, Miami, FL 33125.)

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(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specifications sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The sliding retainer assembly shall be in accordance with D13207E9448 and as specified herein.

3.1.1 Drawing. The drawing forming a part of this specification is an end-product drawing. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the sliding retainer assembly. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3). Any changes or deviations of retainer assemblies from the approved first article during production will be subject to the approval of the contracting officer. Approval of the first article will not relieve the contractor of his obligation to furnish retainer assemblies conforming to this specification.

3.3 Materials. Materials shall be as specified herein and on the drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration and control. The sliding retainer assembly shall be fabricated from compatible materials, inherently corrosion and deterioration resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environments to which the sliding retainer assembly may be exposed.

3.3.2 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the sliding retainer assembly may be newly fabricated from recovered materials to the maximum extent

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practicable, provided the sliding retainer assembly produced meet all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the sliding retainer assembly.

3.4 Welding electrodes. Welding electrodes shall conform to AWS A5.1 or A5.18, classification E60 or E70 series, size as required.

3.5 Identification marking. The sliding retainer assembly shall be metal die stamped as shown on the drawing. Die stamping shall be legible after painting.

3.6 Treatment and painting. The sliding retainer assembly shall be treated and painted in accordance with MIL-T-704, type B.

3.7 Workmanship.

3.7.1 Steel fabrication. Steel used in fabrication shall be free from kinks and sharp bends. The straightening of metal shall be done by methods that will not cause injury to the material. Corners shall be square and true. Flame cutting, using tips suitable for the thickness of the steel, may be employed instead of shearing and sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges and corners, except where sharp edges or corners are required or where they are not detrimental to safety. All bends shall be made to a radius which will not cause cracking at the extreme outer fibers of the piece being bent, and all bends except for minor details shall be made by use of metal dies or fixtures to insure uniformity in size and shape.

3.7.2 Welders and welding.

3.7.2.1 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall obtain and have available the certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the material joined and the type of welding operation to be performed and that such qualification is effective as defined by the particular code:

D1.1 - Structural Welding Code, Section V, Qualification.
Welding Qualifications of the ASME Boiler and Pressure Vessel Code (Section IX).

Contractors who only make horizontal welds need not qualify welders for "all position welding". The contractor shall be responsible for determining that automatic welding equipment operators are capable of producing quality welds in accordance with AWS and ASME codes. The certification shall be made available for review by the contracting officer or the contracting officer's representative.

3.7.2.2 Welding. The surface of parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping and wire brushing, and other foreign matter. Parent materials, weld filler metals, and fabrication techniques shall be as required to enable the sliding retainer

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assembly to conform to the examination and test requirements specified in section 4. Parts to be joined by fillet welds shall be brought into as close contact as possible and in no event shall be separated by more than 1/16 inch unless appropriate bridging techniques are used. Unless otherwise specified (see 6.2), the welding process used in fabrication of the sliding retainer assembly shall be at the option of the contractor.

3.7.3 Assembly. The sliding retainer assembly shall have no misfits or interference of mating parts that will prevent insertion of the retainer bridge pin in the various holes.

3.7.4 Machine work. Tolerances and gages for metal fits shall conform to the limits specified herein and on the applicable drawing and to standards of industry.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Parts and components. Parts and components detailed on the drawing shall be inspected in accordance with the quality assurance provisions (QAP) shown on the drawing. The drawing specifies the characteristics requiring QAP inspection, the sampling plan, and the basis for acceptance and rejection (see 6.4 and 6.5).

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

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4.3 First article inspection.

4.3.1 First article test. The first article shall be examined as specified in 4.5.1, in the order shown. Presence of one or more defects shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined for the major and minor defects specified in 4.5.1. AQL shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.5 Inspection procedure.

4.5.1 Examination. The sliding retainer assembly shall be examined for the following defects:

Major

101. Any part of component not in accordance with the QAP requirements as shown on the drawings.
102. Material not as specified.
103. Components missing or not as specified.
104. Workmanship not as specified.
105. Weld defects such as slag inclusions, spatter, undercutting, porosity, incorrect size, or lack of fusion with parent metal.
106. The bridge pin retainer cannot be inserted in the various holes.
107. Materials are not resistant to corrosion and deterioration or treated to be resistant to corrosion and deterioration for the applicable storage and operating environments.
108. Used, rebuilt, or remanufactured components, pieces or parts incorporated in the retainer assembly.
109. Welders, welding, and welder certification not as specified.
110. Welding electrodes not as specified.

Minor

201. Dimensions other than those identified by QAP data not as specified.
202. Identification marking incorrect, missing, or illegible.
203. Treatment and painting not as specified.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

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4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

111.	Boxes not as specified for level A or B.
112.	Contents not blocked and braced in a manner to prevent free movement for level A or B.
113.	Strapping not as specified for level A.
114.	Marking missing, illegible, incorrect, or incomplete for level A, B, or commercial.

5. PACKAGING

5.1 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

5.1.1 Level A. The sliding retainer assemblies shall be packed together in a close-fitting box conforming to PPP-B-601, overseas type, style optional, or PPP-B-621, class 2, style optional, in quantities not to exceed the weight limitations of the applicable box specification. The contents shall be blocked and braced within the box in a manner to prevent free movement. The box shall be closed and strapped in accordance with the appendix to the applicable box specification. Strapping shall conform to QQ-S-781, class 1, type I or IV, finish B, size as applicable.

5.1.2 Level B. The sliding retainer assemblies shall be packed together as specified in 5.1.1, except the boxes shall be domestic type or class, as applicable. Box closure and strapping shall be in accordance with the appendix to the applicable box specification.

5.1.3 Commercial. The sliding retainer assemblies shall be packed in accordance with ASTM D 3951.

5.2 Marking.

5.2.1 Levels A and B. Marking shall be in accordance with MIL-STD-129.

5.2.2 Commercial. Marking for commercial packaging shall be in accordance with ASTM D 3951. In addition, weight and cube data shall be marked on the shipping container.

6. NOTES

6.1 Intended use. The sliding retainer assembly is used with the universal transom in the construction of fixed and floating bridges when the Class 60 Treadway-Type superstructure is employed.

6.2 Ordering data. Procurement documents should specify the following:

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a. Title, number, and date of this specification.
b. Quantity of sliding retainer assemblies to be furnished for first article examination, when required, and time frame required for submission of the first article (see 3.2).
c. Level of packing required (see 5.1).
d. When metal strapping shall be finish A (see 5.1.1).

6.3 First article. When a first article inspection is required, the item should be a sliding retainer assembly. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests, and approval of the first article test results and disposition of the document's first article.

6.4 Quality assurance provisions (QAP). The contracting officer should require the contractor to maintain records of all QAP inspection. A suggested paragraph is as follows:

"The contractor shall maintain complete records of all examinations and tests performed to verify the requirements of classified QAP characteristics. The records shall include as a minimum, lot size, sample size, drawing requirement, actual measurement, number and type of deficiencies found, quantity approved, quantity rejected, corrected action taken when applicable."

6.5 Definitions.

6.5.1 Quality assurance provisions (QAP). A QAP is a contractual requirement that supplements section 4 of the specification. QAP indicates the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of repair parts.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-R-52224C(ME)	Retainer Assembly, Sliding

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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